HELMHOLTZ ZENTRUM DRESDEN ROSSENDORF

CASUS – Center for Advanced Systems Understanding
Conrad-Schiedt-Straße 20 • D-02826 Görlitz

Computer Physics Communications

CENTER FOR ADVANCED SYSTEMS UNDERSTANDING

CASUS - Center for Advanced Systems Understanding

Dr. Attila Cangi

Head of Department
Machine Learning for Materials Design

Phone +49 3581 - 375 23 52
a.cangi@hzdr.de

29.11.2024

Helmholtz-Zentrum Dresden-Rossendorf e. V.

CASUS – Center for Advanced Systems Understanding

Conrad-Schiedt-Straße 20
D-02826 Görlitz
www.hzdr.de
www.casus.de

Cover Letter

Dear Editors,

We are pleased to submit our manuscript titled "Materials Learning Algorithms (MALA): Scalable Machine Learning for Electronic Structure Calculations in Large-Scale Atomistic Simulations" for consideration as an article in Computer Physics Communications, under the category Computer Programs in Physics (CPiP).

The properties of matter are fundamentally governed by the interactions between electrons and nuclei. Computationally, many of these properties and phenomena can be understood by studying the electronic structure, the probability distribution of electrons in materials or molecules. Efficient methods for modeling the electronic structure have been essential to technological progress.

Despite decades of research, electronic structure simulations with reasonable accuracy remain limited to systems with only a few thousand atoms. This limitation significantly hinders progress, as modeling at larger scales (e.g., systems with more than 100,000 atoms) is essential to solve many complex materials problems.

To address this challenge, we present a computationally efficient approach to overcome this barrier — the MALA package. It combines modern machine learning algorithms with physically motivated input and output quantities to advance electronic structure simulations. By employing spatially resolved machine learning models, our method accurately predicts the electronic structure of materials at scales far beyond what is possible with standard methods such as density functional theory (DFT). MALA therefore enables a leap forward in simulation scale and efficiency, crossing a long-standing frontier in computational physics.

MALA is actively used at institutions such as Sandia National Laboratories, Helmholtz-Zentrum Dresden-Rossendorf (Germany), Georgia Institute of Technology, University of North Carolina A&T, SambaNova Systems, and NVIDIA.

We believe this work will be of great interest to a broad audience spanning computer science, physics, and materials science, making it a strong fit for the excellent portfolio of Computer Physics Communications.

Thank you for considering our manuscript. We look forward to receiving your feedback.

Sincerely,

Attila Cangi